

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

THERMAL PROPERTIES OF MATTER

LECTURE-03 PHYSICS

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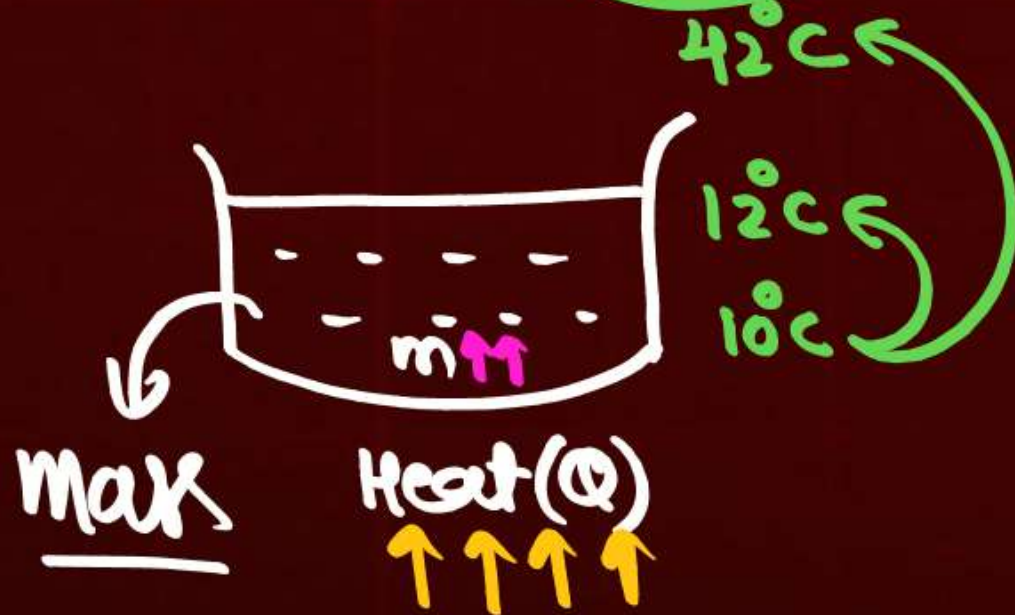
Topics To Be Covered

- 1** Revision
- 2** Specific heat Capacity ✓
- 3** Molar Specific Heat ✓
- 4** Water Equivalent
- 5** Latent Heat Capacity ✓

Heat Capacity

- The amount of heat Required to

↑↑↑ Raise the temperature of a given mass of a Body



lecture. no. 3

Math

$$Q \propto m(\text{mass}) \text{ --- ①}$$

$$Q \propto \Delta T \text{ --- ②}$$

$$Q \propto m \Delta T$$

$$Q = Sm \Delta T$$

$S \Rightarrow$ specific heat capacity



Specific heat capacity S

$$m = 1\text{kg} \quad \Delta T = 1^\circ\text{C}$$

2023
Paper

$$Q = S m \Delta T$$

②

The amount of heat
Required to Raised $\uparrow\uparrow\uparrow$
the temperature
of a substance of
1kg by 1°C

$$\frac{Q}{m \Delta T} = S$$

$$\frac{491}{500}$$

Specific heat capacity

$$S = \frac{Q}{m \Delta T}$$

$\downarrow$ 1kg $\downarrow$ 1°C

Unit of specific heat capacity

Neet-2021

$$S = \frac{\cancel{Q}^{\text{Joule}}}{\cancel{\text{kg}} \cancel{\text{m}} \cancel{\Delta T}^{\text{K}}}$$

$$S = \frac{\text{Joule}}{\text{kg K}}$$

$$S = \frac{\text{Joule}}{\text{kg Kelvin}}$$

Dimensions of (S)

③ 2022 JKROSE

$$S = \frac{Q^{\checkmark}}{\underline{m} \underline{\Delta T}} \quad Q = E = ML^2T^{-2}$$

$$S = \frac{\cancel{M} L^2 T^{-2}}{\cancel{M} \cancel{K}^1}$$

$$S = L^2 T^{-2} K^{-1}$$

$$S = M^0 L^2 T^{-2} K^{-1}$$

Molar Heat Capacity

The amount of heat Q
Required to Raised the
temperature $\uparrow\uparrow\uparrow$
of given no. of moles
of the substance



$$Q \propto n (\text{no. of moles}) \text{ --- ①}$$

$$Q \propto \Delta T \text{ --- II}$$

$$Q \propto n \Delta T$$

$$Q = C n \Delta T$$

$\rightarrow$ Specific
Molar heat
capacity



Specific Molar heat Capacity

$$n = 1 \text{ mole}$$

$$\Delta T = 1^\circ\text{C}$$

$$\underline{1 \text{ mole} = 6.023 \times 10^{23}} \quad \begin{matrix} (2024) \\ 2023 \\ 2025 \end{matrix}$$

ce The amount of heat (Q)
Required to Raised the
temperature of 1 mole
of substance by 1°C

$$Q = \underline{C} n \Delta T$$

$$\frac{Q}{n \Delta T} = \underline{C}$$

$$\underline{C} = \frac{\underline{Q}}{\underset{\substack{\uparrow \\ 1 \text{ mole}}}{\underline{n \Delta T}} \underset{\substack{\uparrow \\ 1^\circ\text{C}}}{\underline{\quad}}}$$

Specific
molar
heat
Capacity

⑤

Unit of C

$$C = \frac{Q}{n \Delta T}$$

$$C = \frac{\text{Joule}}{\text{mole K}}$$

Neet
2019

Dimension of C

$$C = \frac{Q}{n \Delta T} = \frac{H}{n \Delta T} = \frac{ML^2T^{-2}}{\text{mol K}}$$

$$C = \frac{ML^2T^{-2}}{\text{mol K}}$$

$$\Delta T = T_f - T_i$$

$$C = ML^2T^{-2} \text{mol}^{-1} \text{K}^{-1} = 21 - 20 = 1^\circ\text{C}$$

$$\begin{array}{c} 21^\circ\text{C} \\ \uparrow \\ 20^\circ\text{C} \end{array} \quad (6)$$

Note

Thermodynamics

Repeat

$$Q = C n \Delta T$$

Molar heat capacity

$$Q = n C \Delta T \checkmark$$

⑦

Molar heat capacity at Q_p
Constant pressure

$$P_i = 2 \text{ bar} \quad P_f = 2 \text{ bar}$$
$$\Delta P = P_f - P_i$$
$$= 2 - 2 = 0$$



$$Q_p = C_p n \Delta T$$

$$Q_p = n C_p \Delta T$$

Molar heat capacity at constant volume Q_v

$$V_f = 2 \quad V_i = 2$$

$$\Delta V = V_f - V_i$$
$$= 2 - 2 = 0$$

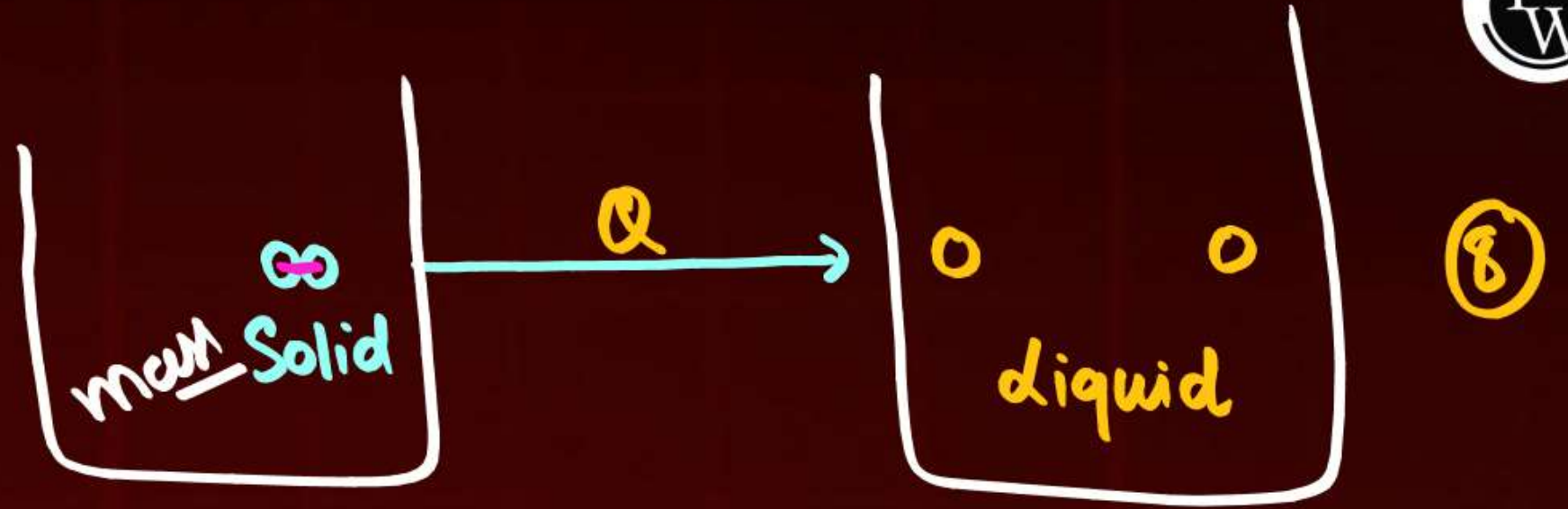


$$Q_v = C_v n \Delta T$$

$$Q_v = n C_v \Delta T$$

Latent Heat

- The heat which is required to change the state of matter
- It helps to break the bond b/w atom
- Latent heat never increases the temperature



$Q \propto m$ (mass of the substance)

$$Q = Lm$$

L → latent heat constant



latent heat

$$Q = Lm$$

(1) latent heat of Fusion Q_f

=> The amount of heat Required to convert solid into liquid



$$Q_f = L_f m$$

(2) latent heat of Vapourization Q_v

=> The amount of heat Required to convert liquid into gas



$$Q_v = L_v m$$

⑨

Calorimetry (Tough)

Heat Energy Measurement

⇒ It is the measurement of all type of heat

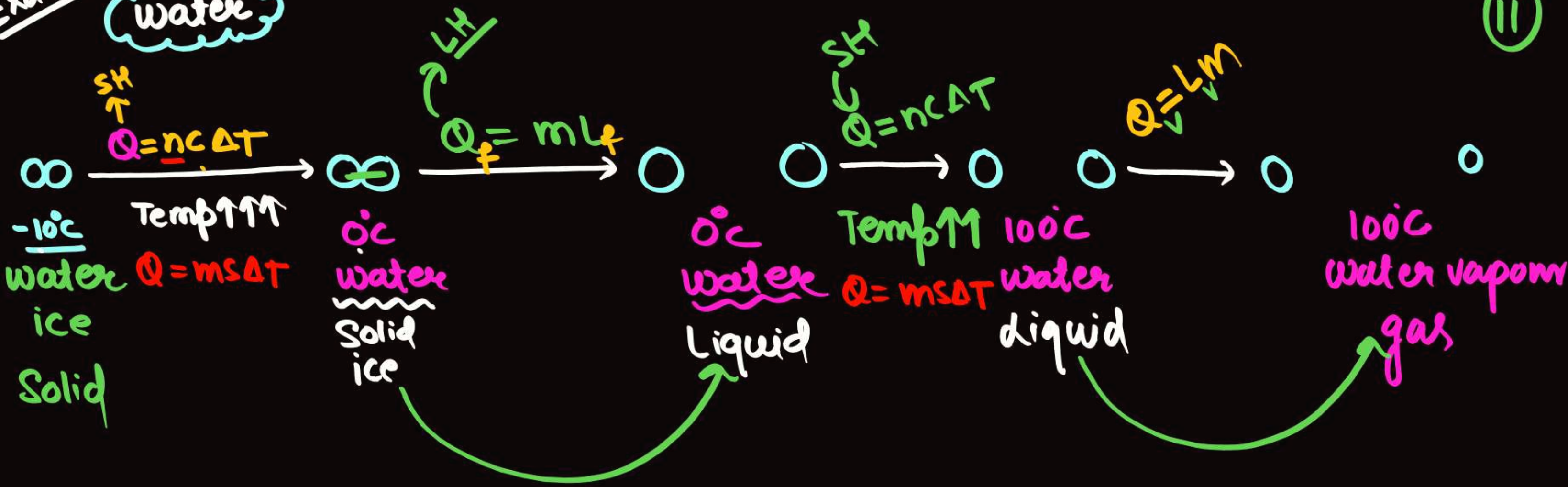
⇒ Heat Capacity (molar) $Q = nCAT$

⇒ latent heat $Q = Lm$



Example

water



Mode of Transfer of Heat

1. Conduction

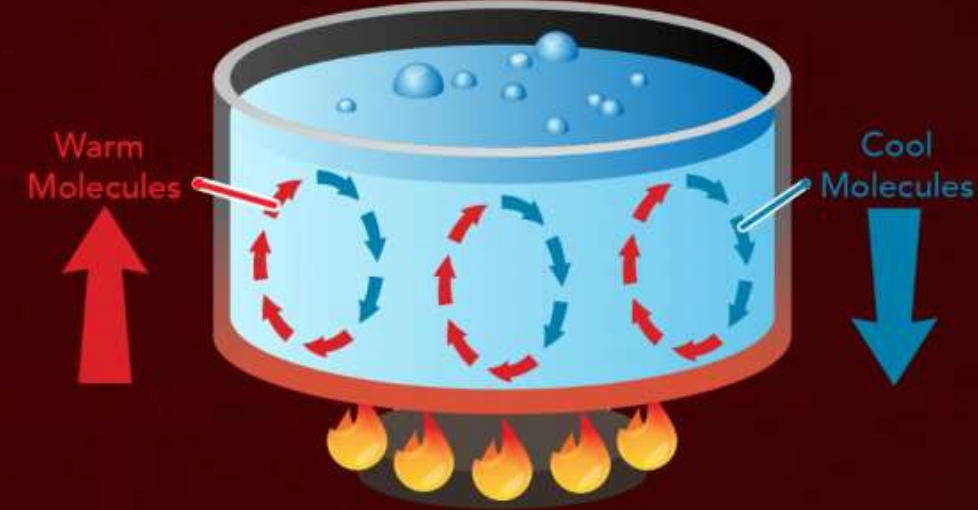
Material particle is required



But particle do not move from one plane to another

2. Convection

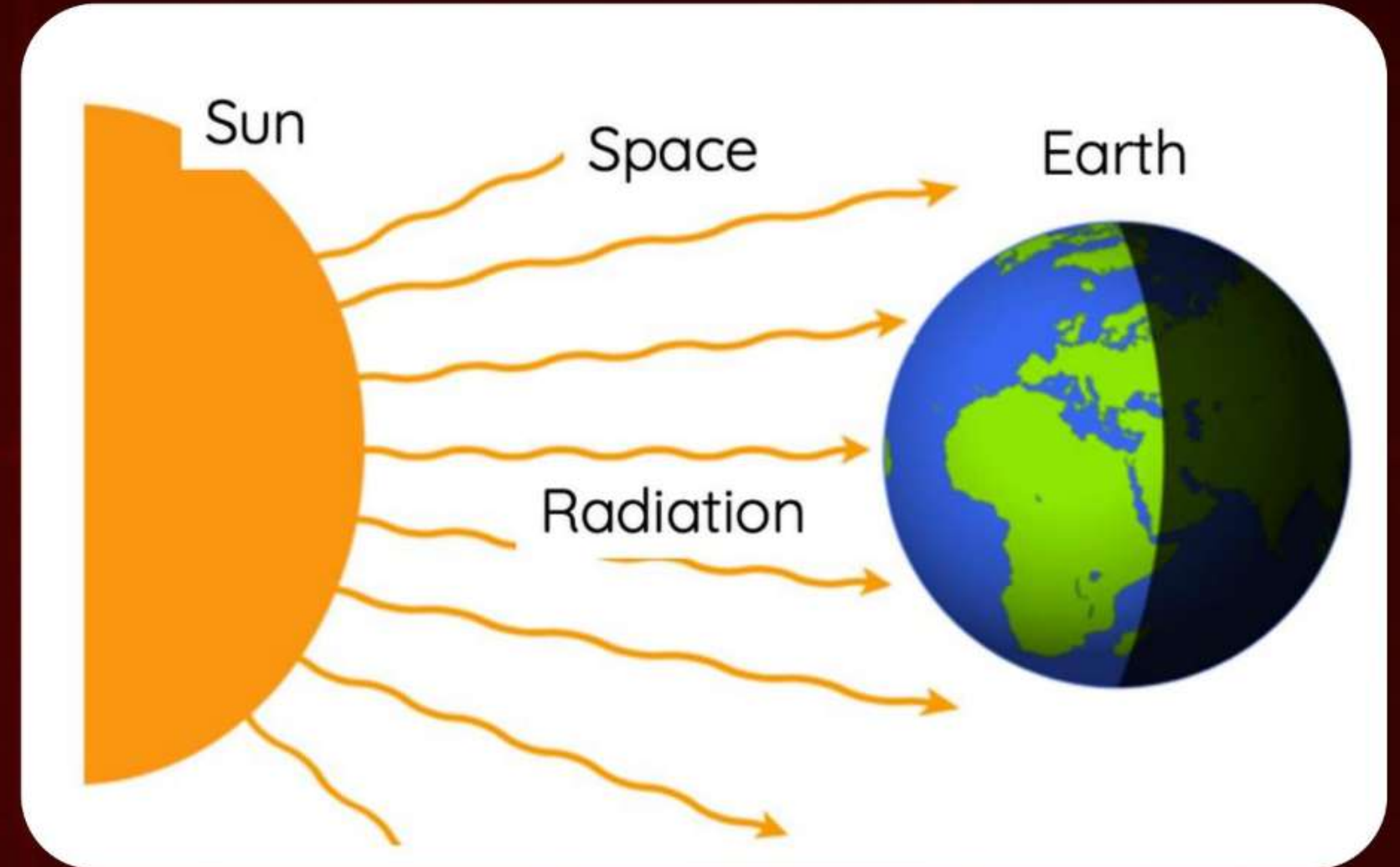
Material particle required



But material particle is moving

3. Radiation

No material medium is required



Conduction

Energy is transferred by direct contact.



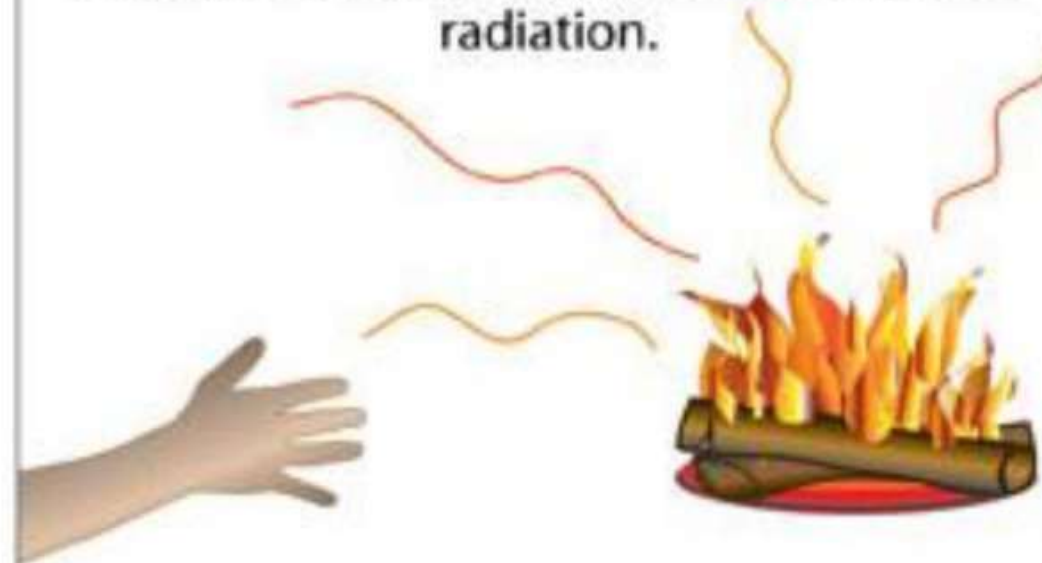
Convection

Energy is transferred by the mass motion of molecules.



Radiation

Energy is transferred by electromagnetic radiation.



SUMMARY

$$Q = mc\Delta T$$

$$Q = nC\Delta T$$

$$Q = mL$$

HOMework



Graph
of (cm)

Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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Thank
You

